Roll No: _____

Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
Auxiliary Examination

BE CSE Third Year
Course Name: Quantum Computing
Time: 3 hours
Marks : 100

Course Code: UCS619
Date: August 19, 2024
Name of Faculty: Dr Samya Muhuri

Note: Attempt all questions with proper justification and working. State your assumptions (if any) and show your working clearly.

1. (a) A system in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle$$

where $\{|u_1\rangle, |u_2\rangle, |u_3\rangle, |u_4\rangle, |u_5\rangle\}$ are complete and orthonormal set of vectors. Each $|u_i\rangle$ is an eigenstate of the system's Hamiltonian corresponding to the possible measurement result $H|u_n\rangle = n\epsilon|u_n\rangle$, where $n = 1, 2, 3, 4, 5$.

Describe the set of projection operators corresponding to the possible measurement results. **(4)**

(b) Determine the probability of obtaining each measurement result. What is the state of the system after measurement if we measure the energy to be 3ϵ? **(4)**

(c) What is the average energy of the system? **(4)**

2. (a) Prove that $Y = iXZ$ using matrix multiplication method. **(5)**

(b) Write down the different steps for creating shifted key based on E91 protocol by considering the entangled pair $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. Explain it with the help of an example. **(7)**

3. (a) A quantum state is in the state

$$|\psi\rangle = \frac{3i|0\rangle + 4|1\rangle}{5}$$

Is the state normalized? Express the state in the $|+\rangle$, $|-\rangle$ basis. **(6)**

(b) Using the appropriate assumptions, describe any three postulates of Quantum Mechanics in a detailed manner. **(6)**

4. (a) Define Quantum Fourier Transform (QFT) for an n-qubit quantum system. **(6)**

(b) Let $n = 1$ qubit, $N = 2$ dimensional space. Apply QFT on the state $\alpha|0\rangle + \beta|1\rangle$. **(6)**

5. Based on the Peter Shor's 9-qubit-code, show the process of rectification of one qubit X-bit error which happened in 2nd qubit.

(12)

6. Consider a function with two inputs such that $f(x) = 1$. Explicitly show that the Deutsch-Jozsa algorithm works in this case by generating the vector $|y\rangle = |00\rangle$ as the final output.

(10)

7. (a) Differentiate between Teleportation and Superdense coding. Describe the procedure by which Alice can send $\alpha|0\rangle + \beta|1\rangle$ to Bob using Teleportation method. **(6)**

   (b) Proof No cloning theorem with any one method. **(4)**

8. (a) Using the matrix representation of the Hadamard gate, write down the matrix $(H \otimes H)(|0\rangle \otimes |1\rangle)$. Show that this is equivalent to

$$|\psi\rangle = (\frac{|0\rangle + |1\rangle}{\sqrt{2}})(\frac{|0\rangle - |1\rangle}{\sqrt{2}})$$

(5)

   (b) Design the circuit for the following entangled quantum systems

$$\frac{1}{\sqrt{2}}(|01100\rangle - |10011\rangle)$$

$$\frac{1}{\sqrt{2}}(|011\rangle - |100\rangle)$$

(5)

9. (a) Write the matrix form of single qubit unitary gate $U(\theta, \phi, \lambda)$. **(4)**

   (b) Put the appropriate values of θ, ϕ, λ to find the matrix of Hadamard gate as

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

(6)

Note- Evaluated answer will be shown on 22nd August, 2024 (Thursday) at L-124, CSED block from 1:30 to 2:30 PM.